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GENOMICS, TRANSCRIPTOMICS, PROTEOMICS

Comparative and analytical characterization of the oral bacteriome of smokeless tobacco users with oral squamous cell carcinoma

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Abstract

Oral cavity squamous cell carcinoma (OSCC) is the most common type of head and neck cancer worldwide. Smokeless tobacco (SLT) has been well proven for its role in oral carcinogenesis due to the abundance of several carcinogens. However, the role of inhabitant microorganisms in the oral cavity of smokeless tobacco users has not yet been well explored in the context of OSCC. Therefore, the present investigation was conceived to analyze the oral bacteriome of smokeless tobacco users having OSCC (CP group). With the assistance of illumina-based sequencing of bacterial-specific V3 hypervariable region of 16S rDNA gene, 71,969 OTUs (operational taxonomic units) were categorized into 18 phyla and 166 genera. The overall analysis revealed that the oral bacteriome of the patients with OSCC, who were smokeless tobacco users, was significantly different compared to the healthy smokeless tobacco users (HTC group) and non-users (HI users). The appearance of 14 significantly abundant genera [FDR (false discovery rate) adjusted probability value of significance (p value) < 0.05] among the CP group showed the prevalence of tobacco-specific nitrosamines forming bacteria (*Staphylococcus*, *Fusobacterium*, and *Campylobacter*). The functional attributes of the oral bacteriome of the CP group can also be correlated with the genes involved in oncogenesis. This study is the first report on the oral bacteriome of Indian patients with OSCC who were chronic tobacco chewers. The results of the present study will pave the way to understand the influence of smokeless tobacco on the oral bacteriome of OSCC patients.

Key points

- *Oral bacteriome of OSCC patients differ from healthy smokeless tobacco (SLT) users and SLT non-users.*
- *Smokeless tobacco influences the oral bacteriome of OSCC group.*
- *Oral bacteriome specific diagnostics may be developed for pre-diagnosis of oral cancer.*

Keywords OSCC; Oral cancer · Smokeless tobacco · Microbiome · *Fusobacterium*

Introduction

South and Southeast Asia populations contribute to the highest percentage of smokeless tobacco users among 127 countries (Siddiqi et al. 2020). Smokeless tobacco (SLT; chewing and unburned form of tobacco) has been linked up with several oral and respiratory diseases including oral cancer, the most prevalent cancer among Asians (Khan et al. 2014). The abundance of toxicants, carcinogens, tobacco-specific nitrosamines (TSNAs), metalloids, heavy metals, polycyclic aromatic hydrocarbons, and volatile aldehydes collectively make tobacco a potentially carcinogenic product (Tomar et al. 2019). However, the role of microorganisms in oral carcinogenesis has not yet been well elucidated. A handful investigations report the dominance of *Acetobacter*, *Bacillus*, *Burkholderia*,

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Corynebacterium, *Staphylococcus*, *Klebsiella*, and *Pseudomonas* like pathogenic bacteria on SLT products (Tyx et al. 2016; Vishwakarma and Verma 2021). Bacterial activity in cured tobacco reduces nitrate into various nitrogen derivatives (NO^{2-}, NO, NO_2, and N_2O_3) which further react with tobacco alkaloids and generate different forms of TSNAs (*N*-nitrosonornicotine (NNN), *N*-nitrosoanatabine (NAT), *N*-nitrosoanabasine (NAB), and 4-(methyl nitrosamino)-1-(3-pyridyl)-1-butanone (NNK)) (Kavvadias et al. 2009; Law et al. 2016). TSNA has been categorized as the group I carcinogen by IARC (International Agency for Research on Cancer) (Lyon 2007). Besides, the abundance of acetaldehyde-forming bacteria (*Rothia*, *Neisseria*, and *Streptococcus* spp.) has also been identified in the oral cavity of SLT users that create a microenvironment for activation of oral oncogenesis (Halboub et al. 2020). The abundance of high carcinogenicity genes in the bacterial composition of SLT samples is a high alarm for tobacco chewers (Al-Hebshi et al. 2017; Srivastav et al. 2020). The inhabitant microorganisms of smokeless tobacco have also been observed to alter the microbiome of the healthy oral cavity (Halboub et al. 2020; Srivastava et al. 2021). Therefore, the effect of SLT on the oral bacteriome among patients with oral squamous cell carcinoma needs to be rigorously studied for identifying significant alterations in bacterial composition. A report elaborates that tumor oral tissues exhibit altered diversity by having more richness as compared to the healthy oral tissues. Here, five species (*Aggregatibacter segnis*, *Capnocytophaga leadbetteri*, *Fusobacterium nucleatum*, *Peptostreptococcus stomatis*, and *Prevotella intermedia*) showed a strong association with tumor tissues. Besides, several genes having a role in carcinogenesis were also significantly abundant in tumor tissues (Zhang et al. 2020).

In our previous investigation, we have observed the influence of smokeless tobacco on the healthy oral bacteriome of smokeless tobacco users, where the elevated counts of pathogens belonging to *Fusobacteria*, *Porphyromonas*, *Desulfobulbus*, *Enterococcus*, and *Parvimonas* were significantly abundant (Srivastava et al. 2021). Here, we have studied the oral bacteriome of OSCC patients (CP group) and also compared their bacterial diversity with the oral bacteriome of healthy smokeless tobacco users (HTC) and non-users (HI) to unravel if any smokeless tobacco-specific bacteriome is associated with the OSCC in the patients. The present investigation elucidates that the oral bacteriome of the patients with OSCC, who were smokeless tobacco users, was significantly different compared to the HTC and HI groups. The investigation paves the way to understand the dynamics of smokeless tobacco on the oral bacteriome of OSCC patients having long-term SLT chewing habits. The present study was conducted in the Indian subcontinent which has a high prevalence of both OSCC patients and SLT users.

Materials and methods

Subject requirement, sample collection, and DNA extraction

Samples were collected from 60 participants from three groups, i.e., HI (20 healthy SLT non-users), HTC (20 healthy SLT users), and CP (19 SLT users who developed OSCC) after ethical approval on the protocol of this study from the Institutional Ethical Committee. Due to the low quality of sequences obtained from CP1, it was not included in the study. Written consent was taken from all the participants of this study after they were informed in detail about the objectives of the study. The CP group participants were regular SLT chewers for more than 5 years and were diagnosed with OSCC (Table 1). Those who smoked tobacco, consumed alcohol, and received antibiotic medication for the last 3 months were excluded from the study. Here, the NGS (next-generation sequencing) sequences of the HI group (Verma et al. 2021) and HTC group (Srivastava et al. 2021) have been used in this investigation as a control for comparing with the sequences of the CP group subjects. The HI and HTC groups comprise 20 members of the same demographic region in each category. The HI group (age 21–57 years) includes healthy participants that had never consumed SLT in any form, while the HTC group (age 24–60 years) comprises healthy members having SLT chewing habits for the last 5 years. Sterile saline water (40 ml) was provided to the participants for gargling; the oral wash was collected into a 50-ml sterile tube. The samples so obtained from the participants were centrifuged for bacterial pellet collection followed by DNA extraction. Metagenomic DNA was extracted by using the Qiagen DNeasy Blood and Tissue Kit (Qiagen, Hilden, Germany) after the prior treatment of pellet with proteinase-K (10 mg/ml) and lysozyme (10 mg/ml). The amplicon library of the V3 region was generated with 5 ng of an extracted metagenome using Ex*Taq* polymerase of TAKARA (DSS Takara Bio, New Delhi, India).

Illumina sequencing, sequence processing, and taxonomy assignment

The amplicon (150 bp) was sequenced at the Illumina MiSeq2500 platform (Illumina, San Diego, USA). The sequences were submitted to Sequence Read Archive (SRA) with the Bioproject number (PRJNA756784). The FastQC analysis provided sequences of Phred value > 30 that were processed for obtaining quality reads using different filters such as conserved region, spacer, mismatch, and consensus

Table 1 Various parameters of the OSCC suffered smokeless tobacco users during this investigation

S. no	Code	Age	Gender	Food habit	Area affected	Tobacco chewing habit	Family history	Duration of chewing (years)	Diabetic status[$]	Systolic BP[#]	BMI (Body mass index)[δ]
1	CP2	40	Female	Vegetarian	Tongue left lateral	Loose leaves and pan masala	No	07	No	120	24.7
2	CP3	45	Male	Non-vegetarian	Middle of tongue	Loose leaves and $CaCO_3$	No	30	Yes	130	28.9
3	CP4	44	Male	Non-vegetarian	Right cheek	Pan masala	Elder brother	04	No	140	24.5
4	CP5	63	Male	Non-vegetarian	Left check and gums	Loose leaves and pan masala	No	20	Yes	134	26.8
5	CP6	35	Male	Non-vegetarian	Left cheek	Gutkha	No	10	No	124	32
6	CP7	63	Male	Vegetarian	Left cheek	Loose leaves and gutkha	No	10	No	118	22
7	CP8	52	Male	Vegetarian	Tongue	Tobacco	No	10	No	112	28.8
8	CP9	56	Male	Vegetarian	Tongue cancer	Loose leaves and gutkha	No	10	No	126	24.6
9	CP10	60	Male	Non-vegetarian	Deep inside the teeth	Surti alone	No	05	No	122	30
10	CP11	50	Male	Non-vegetarian	Right cheek	Surti and $CaCO_3$	No	10	No	136	24.5
11	CP12	65	Male	Non-vegetarian	Right cheek	Surti and $CaCO_3$	No	40	Yes	142	27.8
12	CP13	53	Female	Non-vegetarian	Supragingival epithelial squamous	Loose leaves and gutkha	No	05	No	116	22.9
13	CP14	48	Male	Non-vegetarian	Squamous epithelium hyperplasia	Loose leaves and gutkha	No	06	No	132	23.7
14	CP15	27	Male	Non-vegetarian	OSCC	Loose leaves and gutkha	No	05	No	124	24.8
15	CP16	50	Male	Non-vegetarian	Left cheek	Leaves, khaini, and gutkha	No	05	No	120	27.4
16	CP17	46	Male	Non-vegetarian	Lateral tongue	Leaves, khaini, and gutkha	No	05	No	128	28.7
17	CP18	52	Female	Non-vegetarian	Right cheek OSCC	Leaves and khaini	No	05	No	118	22.3
18	CP19	56	Female	Vegetarian	Left cheek	Leaves and khaini	No	05	No	120	21.8
19	CP20	33	Male	Non-vegetarian	Left cheek	Leaves and khaini	No	06	No	130	26.7

Samples of HI and HTC groups were collected from the same demographic region. The other parameters of the healthy SLT users (HTC) non-users (HI) are mentioned in Table 1 (Srivastava et al. 2021) and supplementary Table 1 (Verma et al. 2021) of respective publications. Khaini: cured tobacco cut leaves mixed with slaked lime; Gutkha: dried, finely chopped tobacco mixed with areca nuts, flavors, and catechu and slaked lime; Surti: local name of sun-dried tobacco leaves of India and Nepal regions

S. no. serial number

[$]All the participants of HI group and HTC groups (except one) were non-diabetic (Verma et al. 2021)

[#]Systolic BP HI group were in the range of 110–145 while HTC group showed systolic BP in the range of 112–146

[δ]BMI of HI group were in the range of 20–28 while HTC group showed BMI in the range of 21.6–29.1

reads. Various bioinformatic tools ClustalO program (Sievers et al. 2011) and UCHIME (Edgar et al. 2011) of USEARCH (Edgar 2013), UCLUST (similarity cutoff = 0.97) were used at various steps of downstream processing of the sequences at QIIME 1.9.1 (Caporaso et al. 2010). The Greengenes database (https://greengenes.secondgenome.com/) was used as reference data to align the representative sequences (DeSantis et al. 2006a) and Taxonomy assignment using the PyNAST program (DeSantis et al. 2006b).

Data interpretation and statistical analysis

Bacterial composition was analyzed for various taxon levels with the main emphasis on phyla- and genera-level analysis. Diversity indices (Chao1, Shannon, Simpson, PD whole_tree, and Good's estimator) were calculated (Good 1953; Bik et al. 2010). Unknown sequences are defined as unaligned sequences of taxonomic databases. Comparative analysis among the HI, HTC, and CP groups has been carried out to determine diversity indices as well as significantly different [false discovery rate (FDR) adjusted p value < 0.05] phyla and genera. The data were checked for normality, and in case of violation of normality assumption, Mann Whitney U test was applied to calculate the p values at the phylum and genera level. The principal coordinate analysis (PCoA) plots were produced using the first two principal coordinates and labeled according to OSCC status using the Bray–Curtis distance matrix. The functional characteristics were predicted using the PICRUSt (phylogenetic investigation of communities by reconstruction of unobserved states) 1.1.0 analysis tool (Langille et al. 2013; https://github.com/picrust/) by associating the 16S rRNA sequences with the KEGG database (https://www.genome.jp/kegg/). Thus, obtained functions were compared between the groups (CP and HTC) to determine the statistically significant functional attributes using the STAMP tool (Parks et al. 2014) where relevant characters were analyzed by using a two-sided G test (w/ Yates) + Fisher's test at a 95% confidence interval. Supplementary files 1 and 2 (HTC; Srivastava et al. 2021) and supplementary files 3 and 4 (HI; Verma et al. 2021) were used to compare the taxa among all three groups.

Results

FastQC analysis and pre-processing of the reads

A total of 11,783,103 raw reads were converted into 71,969 OTUs after filtering and singleton removal to get good-quality reads for taxonomy assignment from 19 participants of CP group. The summary of the reads after each filter is tabulated (Supplemental Table S1 to S4). The generation of such a high number of OTUs may be due to the QIIME 1.9.1-UCLUST that produces a large number of counterfeit OTUs (Prodan et al. 2020). However, it is a routine procedure of microbiome analysis where UCLUST offers the fastest run-time along with good performance (Bokulich et al. 2018). The information of raw reads and total OTUs of HI and HTC groups are summarized in Supplemental Table S5.

Dominant phyla and genera

Of the CP group, a total of 18 phyla, 32 classes, 63 orders, 111 families, and 166 genera were assigned from 71,969 OTUs that were obtained after filtration and the removal of singletons. A significant portion of OTUs (26.27%) was identified as unassigned that did not categorize into any phylum, and 30.81% of OTUs could not be categorized into any genus. Six dominant phyla showed their prominence more than 1% that include *Firmicutes* (24.344%), *Bacteroidetes* (21.31%), *Proteobacteria* (17.68%), *Fusobacteria* (4.52%), *Actinobacteria* (3.54%), and *Candidate division _GN02* (1.27%) (Fig. 1A). Similarly, of the total 166 genera, *Prevotella* (9.17%), *Neisseria* (8.12%), *Hemophilus*

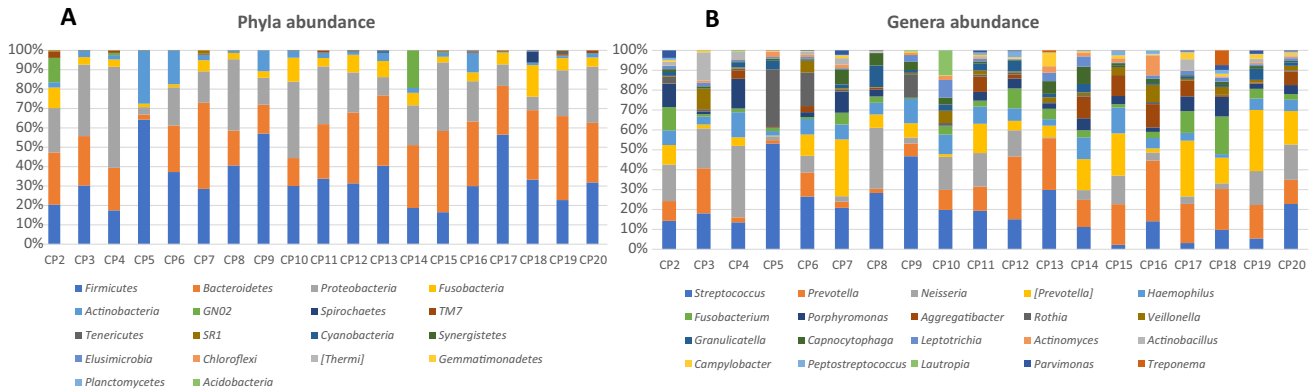


Fig. 1 The relative abundance (%) of phyla (A) and top 20 genera (B) present in the oral cavity of each participant of CP group

(4.42%), *Porphyromonas* (3.07%), *Fusobacterium* (3.02%), *Agrigatibacter* (2.19%), *Rothia* (1.98%), *Veillonella* (1.85%), *Granulicatella* (1.71%), *Capnocytophega* (1.58%), *Actinobacillus* (1.21%), *Leptotrichia* (1.18%), and *Actinomyces* (1.03%) showed their relative abundance > 1% (Fig. 1B). The calculated diversity indices Simpson (SI), Shannon (SH), and Chao1 (Ch) were highest in the subject CP-18 (SI index 0.97; SH index 7.09; Ch index 9332.96), while CP-5 showed the least bacterial diversity for all the three diversity indices (SI index 0.89; SH index 4.46; Ch index 4410.19). Further, CP-15 exhibited the second highest diversity (SI index 0.97; SH index 6.42) among all the participants, whereas CP-18 showed a maximum Chao 1 index (9332.96) (Table 2). Comparative analysis among HI, HTC, and CP groups was further analyzed for various diversity indices. Chao1 (species richness) was found to be highest in HI followed by the HTC and CP groups; however, the CP group showed higher Shannon and Simpson diversity indices followed by the HTC group and least by the HI group (Fig. 2).

Shared phyla and genera of the CP group

Of the total 18 bacterial phyla, seven different phyla (*Firmicutes*, *Bacteroidetes*, *Proteobacteria*, *Fusobacteria*, *Actinobacteria*, *Spirochaetes*, and *Candidate division_TM7*), were observed among all the participants (Supplemental Table S6). Phylum *Synergistes* showed presence among 18 participants (~95%). *Candidate division_GN02*, *Tenericutes*, and *Chloroflexi* were shared by 89% of the participants. The *Candidate division_SR1*

Table 2 Diversity indices and estimators obtained among OSCC suffered SLT users

Sample ID	OTU observed		Dominant phyla (%)	Dominant genera (%)	Chao1	Shannon	Simpson	PD_whole tree
	Phyla level	**Genus level**						
CP-2	15	129	*Bacteroidetes* (21.94)	*Neisseria* (11.62)	5547.039	6.175662	0.965983	137.1859
CP-3	13	124	*Proteobacteria* (30.53)	*Prevotella* (17.51)	4593.34	5.760151	0.960421	114.0131
CP-4	14	141	*Proteobacteria* (38.89)	*Neisseria* (24.29)	4526.969	5.604667	0.948843	106.4883
CP-5	17	181	*Firmicutes* (35.80)	*Streptococcus* (22.33)	4410.193	4.460586	0.896595	110.2667
CP-6	13	114	*Firmicutes* (24.72)	*Streptococcus* (16.22)	5178.924	5.537131	0.936169	113.9292
CP-7	15	144	*Bacteroidetes* (31.39)	*[Prevotella]* (17.93)	5588.659	5.632873	0.940881	136.4597
CP-8	16	154	*Firmicutes* (29.66)	*Neisseria* (21.27)	4871.4	5.144591	0.92892	119.0845
CP-9	12	121	*Firmicutes* (39.48)	*Streptococcus* (26.79)	4991.446	4.911266	0.918519	113.7232
CP-10	13	130	*Proteobacteria* (24.43)	*Streptococcus* (11.46)	5865.71	5.662228	0.945024	134.6718
CP-11	15	160	*Firmicutes* (25.63)	*Streptococcus* (12.58)	8579.332	6.537109	0.960751	176.6283
CP-12	15	178	*Bacteroidetes* (29.17)	*Streptococcus* (10.47)	7167.019	5.917604	0.957665	166.9239
CP-13	11	125	*Firmicutes* (26.29)	*Streptococcus* (16.83)	5531.633	5.424224	0.925984	129.7988
CP-14	11	121	*Bacteroidetes* (20.53)	*[Prevotella]* (7.23)	5883.471	5.383614	0.923527	132.0461
CP-15	13	173	*Bacteroidetes* (35.54)	*[Prevotella]* (15.79)	7667.88	6.421354	0.971424	189.9864
CP-16	13	143	*Bacteroidetes* (27.79)	*Prevotella* (22.94)	6916.748	6.233732	0.96874	160.4357
CP-17	14	139	*Firmicutes* (41.07)	*[Prevotella]* (9.02)	5298.03	5.030724	0.899115	125.8816
CP-18	15	190	*Bacteroidetes* (30.58)	*Prevotella* (13.14)	9332.967	7.094967	0.974258	245.5919
CP-19	14	158	*Bacteroidetes* (34.75)	*[Prevotella]* (20.63)	8615.952	6.253174	0.962264	195.8584
CP-20	14	170	*Firmicutes* (23.68)	*Streptococcus* (14.98)	9054.192	5.991696	0.948879	181.2766

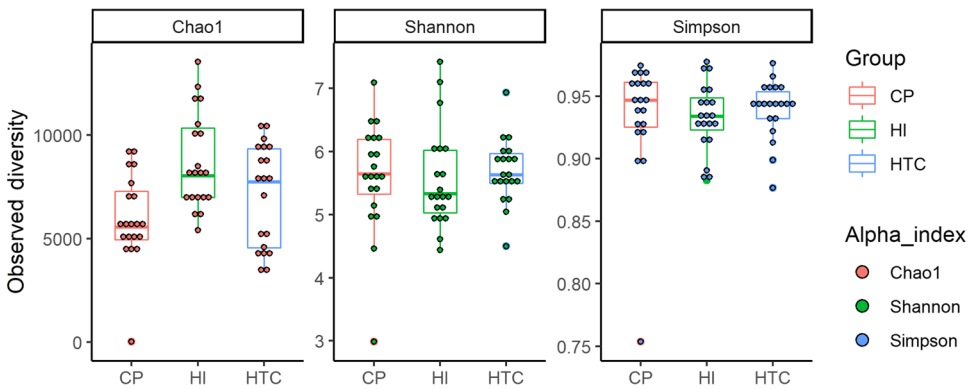


Fig. 2 Box plot analysis showing the alpha diversity indices (Chao1, Shannon, and Simpson index) in bacterial microbiomes among CP, HI, and HTC groups. Median values and inter-quartile ranges have been indicated in the plots

was present in 16 participants (84%). *Chloroflexi* (~ 32%), *Thermi* (15%), and *Elusimicrobia* (10%) were observed in less than 50% of the population, while phyla *Gemmatimonadetes, Planctomycetes*, and *Acidobacteria* showed their presence in only one participant.

Of the 166 assigned genera, 34 genera were shared among the oral cavities of all the participants (Supplemental Table S7), where the top 10 genera include *Streptococcus*, *Prevotella*, *Neisseria*, *Haemophilus*, *Fusobacterium*, *Porphyromonas*, *Aggregatibacter*, *Rothia*, *Veillonella*, *Granulicatella*, and *Capnocytophaga.* Nine different genera, *Peptostreptococcus*, *Pseudomonas*, *Gordonia*, *Bacteroides*, *Microbacterium*, *Atopobium*, *Brevibacterium*, *Paenibacillus*, and *Eikenella* were present in 18 participants (94%). Four genera (*Mycoplasma*, *Lactobacillus*, *Tannerella*, and *Syntrophomonas*) were observed among 89% of the participants. *Paludibacter*, *Bifidobacterium*, and *Pyramidobacter* were present in 84% of the participants.

Statistical analysis

Statistical analysis revealed three significantly different phyla among the three groups (HI, HTC, and CP) that included *Fusobacteria* (*p* value 0.01), *Spirochaetes* (*p* value 0.018), and *Chloroflexi* (*p* value 0.005) (Fig. 3A), while 14 genera (*Campylobacter*, *Capnocytophaga*, *Cotonella*, *Coprococcus*, *Corynebacterium*, *Eikenella*, *Enterococcus*, *Fusobacterium*, *Gordonia*, *Lactobacillus*, *Microbacterium*, *Moryella*, *Oribacterium*, and *Staphylococcus*) had been detected as significantly different by considering FDR adjusted *p* values among the three groups, i.e., HI, HTC, and CP (Fig. 3B). However, on the scale of the *p* values, 26 genera had been observed as significantly different (Table 3). We have also attempted phyla-level statistical analysis among CP patients at the gender level (Supplemental Table S8), where the phylum *Fusobacteria* was observed as significantly different. However, substantial differences in sample size at the gender level (15:4 male/female) cause limitations in the interpretation of such results.

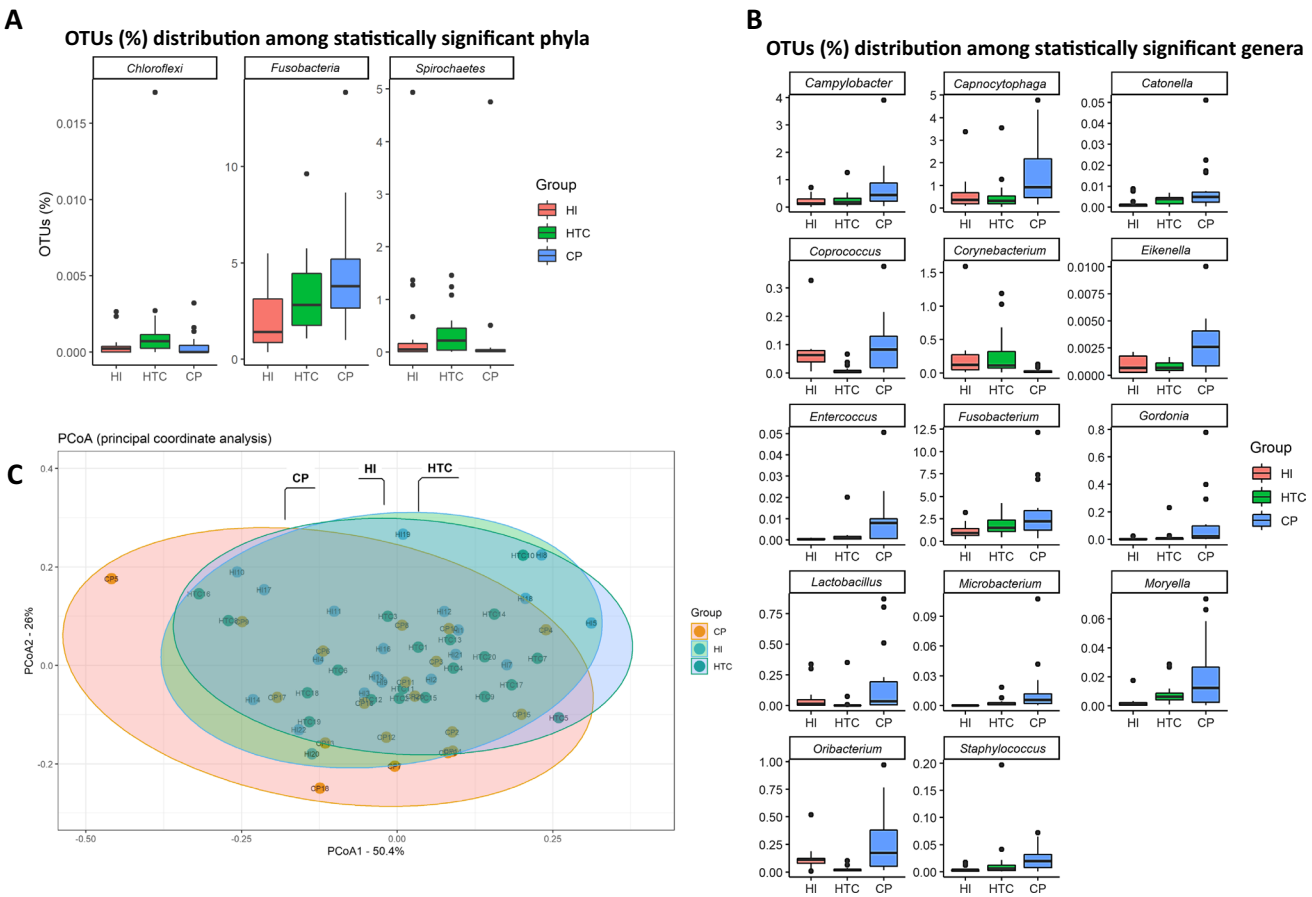


Fig. 3 The OTU (%) distribution of statistically significant phyla (A) and genera (B) among HI, HTC, and CP group participants. The PCoA analysis was conducted using Bray–Curtis distance which was plotted in R software using ggplot2 (C). The ellipse was plotted; ellipses are estimated using the Khachiyan algorithm

Table 3 Analysis to determine the statistically significant genera between HI, HTC, and CP groups

Genera	Comparison of genera across the three groups			Statistical significance (p value)	Hochberg-FDR (p value)
	Mean ± SD (n)				
	CP	**HTC**	**HI**		
Prevotella	7.0715 ± 5.8534 (19)	4.9629 ± 5.6194 (20)	3.8468 ± 3.8677 (20)	0.114	0.303
Acholeplasma	0.0009 ± 0.0004 (2)	0.0104 ± 0.0193 (4)	0.0053 ± 0.0041 (3)	0.705	0.832
Acinetobacter	0.0226 ± 0.0704 (11)	0.169 ± 0.3781 (6)	0.0006 ± 0.0005 (10)	0.121	0.305
Actinobacillus	1.2115 ± 2.4943 (19)	2.8311 ± 2.8577 (20)	1.2182 ± 1.4444 (20)	0.044	0.171
Actinomyces	1.0312 ± 1.6544 (19)	0.7967 ± 0.7933 (20)	1.2798 ± 1.4603 (20)	0.562	0.745
Aggregatibacter	2.1963 ± 2.7827 (19)	0.566 ± 0.4654 (20)	0.582 ± 0.7418 (20)	0.420	0.633
Anaerovorax	0.0009 ± 0.0008 (5)	0.0035 ± 0.0039 (9)	0.007 ± 0.0075 (7)	0.016	0.101
Arcobacter	0.0025 ± 0.0026 (7)	0.0005 ± 0.0001 (2)	0.0003 ± 0.0003 (8)	0.009	0.061
Arenibacter	0.0012 ± 0.0004 (2)	0.0014 ± 0 (1)	0.0007 ± 0.0007 (3)	0.807	0.858
Atopobium	0.007 ± 0.0079 (18)	0.0222 ± 0.0374 (20)	0.0045 ± 0.0033 (18)	0.194	0.4
Bacillus	0.0009 ± 0.0012 (15)	0.0009 ± 0.0015 (11)	0.0008 ± 0.0008 (14)	0.305	0.505
Bacteroides	0.0282 ± 0.0708 (18)	0.0299 ± 0.0393 (19)	0.0114 ± 0.0111 (20)	0.044	0.171
BF311	0.0012 ± 0.001 (14)	0.0035 ± 0.008 (12)	0.0051 ± 0.0078 (14)	0.048	0.18
Bifidobacterium	0.0168 ± 0.0201 (16)	0.0327 ± 0.0906 (17)	0.014 ± 0.0248 (6)	0.515	0.704
Bilophila	0.0151 ± 0 (1)	0.0027 ± 0 (1)	0.0023 ± 0.0029 (2)	0.407	0.623
Brevibacterium	0.0065 ± 0.0085 (18)	0.0029 ± 0.0069 (14)	0.0015 ± 0.0022 (15)	0.031	0.149
Bulleidia	0.2014 ± 0.5896 (19)	0.0233 ± 0.0162 (20)	0.0218 ± 0.019 (20)	0.251	0.447
Butyrivibrio	0.0075 ± 0.0142 (19)	0.0052 ± 0.0049 (20)	0.0056 ± 0.0061 (19)	0.516	0.704
Campylobacter	0.7267 ± 0.866 (19)	0.2716 ± 0.2702 (20)	0.2286 ± 0.204 (20)	0.004	0.034
Candidatus.Tammella	0.0004 ± 0.0002 (2)	0.0097 ± 0.0124 (2)	0.001 ± 0.0004 (2)	0.156	0.352
Capnocytophaga	1.581 ± 1.5127 (19)	0.562 ± 0.7679 (20)	0.7166 ± 0.9608 (20)	0.004	0.034
Cardiobacterium	0.0091 ± 0.0084 (12)	0.0037 ± 0.0038 (19)	0.0078 ± 0.0094 (19)	0.240	0.447
Catonella	0.0087 ± 0.012 (19)	0.0033 ± 0.0019 (19)	0.0018 ± 0.0026 (17)	0.002	0.022
Cetobacterium	0.0011 ± 0.0013 (5)	0.0145 ± 0.0208 (10)	0.0021 ± 0.0038 (11)	0.198	0.4
Cloacibacterium	0.0009 ± 0.0011 (10)	0.0007 ± 0.0005 (8)	0.0004 ± 0.0001 (4)	0.333	0.542
Collinsella	0.0042 ± 0.0042 (13)	0.0022 ± 0.0026 (12)	0.001 ± 0.0007 (13)	0.039	0.164
Coprococcus	0.0959 ± 0.0949 (19)	0.0109 ± 0.0164 (20)	0.0676 ± 0.0664 (20)	0.000	0
Corynebacterium	0.0337 ± 0.0377 (19)	0.2699 ± 0.3309 (20)	0.2112 ± 0.3428 (20)	0.000	0
Desulfobulbus	0.0039 ± 0.0052 (2)	0.0026 ± 0.0029 (14)	0.0011 ± 0.0013 (11)	0.029	0.146
Desulfomicrobium	0.0007 ± 0.0008 (3)	0.0112 ± 0 (1)	0.0189 ± 0 (1)	0.202	0.4
Desulfosporosinus	0.0005 ± 0.0003 (6)	0.0018 ± 0.0027 (3)	0.0004 ± 0.0002 (4)	0.756	0.857
Desulfovibrio	0.1385 ± 0.4129 (9)	0.0034 ± 0.0034 (6)	0.0052 ± 0.0063 (9)	0.402	0.623
Dethiosulfatibacter	0.0006 ± 0.0005 (6)	0.0007 ± 0.0004 (5)	0.0005 ± 0.0003 (5)	0.797	0.858
Dialister	0.148 ± 0.2087 (19)	0.0694 ± 0.0582 (20)	0.0595 ± 0.0601 (20)	0.801	0.858
Eikenella	0.0029 ± 0.0025 (18)	0.0009 ± 0.0005 (12)	0.001 ± 0.0008 (13)	0.008	0.058
Enhydrobacter	0.0007 ± 0.0004 (3)	0.4937 ± 1.5964 (11)	0.0094 ± 0.0274 (13)	0.703	0.832
Enterococcus	0.0096 ± 0.0134 (15)	0.0025 ± 0.0054 (13)	0.0005 ± 0.0004 (11)	0.003	0.03
Epulopiscium	0.0005 ± 0.0003 (2)	0.0005 ± 0.0003 (5)	0.0008 ± 0.0007 (4)	0.861	0.887
Erwinia	0.0004 ± 0.0002 (2)	0.0513 ± 0.0884 (3)	0.0005 ± 0 (1)	0.667	0.815
Filifactor	0.1327 ± 0.2809 (14)	0.1082 ± 0.1176 (14)	0.0886 ± 0.1549 (18)	0.131	0.315
Fusibacter	0.0003 ± 0.0001 (2)	0.001 ± 0.0013 (4)	0.0015 ± 0.0016 (4)	0.073	0.217
Fusobacterium	3.0204 ± 2.9196 (19)	1.8631 ± 1.1458 (20)	1.1474 ± 0.8007 (20)	0.007	0.054
Gemella	0.2661 ± 0.3793 (19)	0.1219 ± 0.1163 (20)	0.1247 ± 0.1212 (20)	0.573	0.745
Gordonia	0.1082 ± 0.1993 (18)	0.0229 ± 0.0603 (14)	0.0047 ± 0.0084 (16)	0.000	0
Granulicatella	1.7157 ± 1.7211 (19)	2.9778 ± 3.9024 (20)	1.8962 ± 2.3454 (20)	0.650	0.81
Haemophilus	4.4201 ± 2.3223 (19)	4.4028 ± 3.2304 (20)	8.1937 ± 7.3411 (20)	0.119	0.305
Jeotgalicoccus	0.0013 ± 0.0011 (7)	0.0008 ± 0.001 (10)	0.0014 ± 0.0017 (10)	0.346	0.555

Table 3 (continued)

Genera	Comparison of genera across the three groups			Statistical significance (p value)	Hochberg-FDR (p value)
	Mean ± SD (n)				
	CP	**HTC**	**HI**		
Kingella	0.0094 ± 0.0078 (19)	0.0142 ± 0.0221 (20)	0.0432 ± 0.0834 (20)	0.471	0.673
Klebsiella	0.0131 ± 0.0415 (11)	0.9236 ± 3.3215 (13)	0.0004 ± 0.0002 (11)	0.250	0.447
Lactobacillus	0.1788 ± 0.2799 (17)	0.0277 ± 0.0881 (16)	0.0577 ± 0.105 (16)	0.002	0.022
Lactococcus	0.0005 ± 0.0002 (5)	0.0047 ± 0.0091 (5)	0.001 ± 0.0011 (4)	0.373	0.589
Lautropia	0.4443 ± 1.6395 (19)	0.1118 ± 0.1449 (20)	0.0789 ± 0.0905 (20)	0.710	0.832
Leptotrichia	1.1812 ± 1.1857 (19)	0.6948 ± 0.6398 (20)	0.9423 ± 1.0444 (20)	0.248	0.447
Megasphaera	0.1035 ± 0.1501 (19)	0.044 ± 0.0544 (20)	0.1111 ± 0.177 (20)	0.572	0.745
Microbacterium	0.0146 ± 0.0255 (18)	0.003 ± 0.0042 (19)	0.0003 ± 0.0001 (3)	0.002	0.022
Micrococcus	0.0021 ± 0.0034 (11)	0.0011 ± 0.0011 (7)	0.0005 ± 0.0002 (16)	0.108	0.295
Mogibacterium	0.2063 ± 0.2554 (19)	0.166 ± 0.1673 (20)	0.1394 ± 0.2044 (20)	0.265	0.454
Moraxella	0.0024 ± 0.0041 (15)	0.1649 ± 0.4231 (15)	0.8554 ± 2.1544 (18)	0.751	0.857
Moryella	0.0209 ± 0.0234 (19)	0.0081 ± 0.0074 (20)	0.0021 ± 0.0038 (20)	0.000	0
Mycobacterium	0.0007 ± 0.0005 (3)	0.0015 ± 0.0019 (4)	0.0029 ± 0.0026 (16)	0.067	0.211
Mycoplasma	0.1778 ± 0.3405 (17)	0.0736 ± 0.0856 (20)	0.0588 ± 0.0861 (16)	0.135	0.317
Neisseria	8.1276 ± 7.0553 (19)	14.3078 ± 8.8658 (20)	13.1137 ± 12.1808 (20)	0.056	0.195
Oribacterium	0.2553 ± 0.2578 (19)	0.0249 ± 0.0232 (20)	0.1192 ± 0.1042 (20)	0.000	0
Paenibacillus	0.0053 ± 0.0049 (18)	0.0057 ± 0.0114 (18)	0.0037 ± 0.0056 (17)	0.177	0.389
Paludibacter	0.0409 ± 0.0572 (16)	0.025 ± 0.0429 (20)	0.0333 ± 0.0368 (19)	0.511	0.704
Parvimonas	0.4505 ± 0.6802 (19)	0.2409 ± 0.3905 (20)	0.0873 ± 0.1554 (20)	0.053	0.191
Pelotomaculum	0.0011 ± 0.0013 (6)	0.0007 ± 0.0008 (2)	0.0009 ± 0.0009 (5)	0.772	0.857
Peptococcus	0.0143 ± 0.0186 (13)	0.0119 ± 0.0098 (16)	0.0212 ± 0.0283 (20)	0.472	0.673
Peptoniphilus	0.0094 ± 0.0121 (12)	0.0059 ± 0.0142 (9)	0.004 ± 0.0086 (8)	0.036	0.158
Peptostreptococcus	0.4689 ± 0.6256 (18)	0.1931 ± 0.4531 (19)	0.2418 ± 0.7124 (20)	0.124	0.305
ph2	0.0027 ± 0.0048 (4)	0.0021 ± 0 (1)	0.0061 ± 0 (1)	0.607	0.776
Porphyromonas	3.0765 ± 2.7053 (19)	4.2508 ± 3.7316 (20)	2.4778 ± 2.2271 (20)	0.157	0.352
Prevotella.1	9.1752 ± 6.7492 (19)	9.4931 ± 6.2723 (20)	9.382 ± 6.5944 (20)	0.898	0.914
Propionibacterium	0.0032 ± 0.0078 (13)	0.002 ± 0.0034 (12)	0.0007 ± 0.0005 (14)	0.670	0.815
Propionigenium	0.0003 ± 0.0001 (2)	0.0003 ± 0.0002 (4)	0.0004 ± 0.0002 (5)	0.434	0.645
Propionivibrio	0.0004 ± 0.0002 (2)	0.0003 ± 0.0002 (6)	0.0007 ± 0 (1)	0.252	0.447
Pseudomonas	0.1713 ± 0.4099 (18)	0.0035 ± 0.0077 (10)	0.0006 ± 0.0005 (8)	0.018	0.107
Pyramidobacter	0.0064 ± 0.0115 (16)	0.0306 ± 0.0763 (8)	0.0018 ± 0.0017 (7)	0.196	0.4
RFN20	0.0112 ± 0.0128 (4)	1.5798 ± 1.0072 (6)	0.0006 ± 0.0003 (3)	0.019	0.107
Rothia	1.9854 ± 3.5985 (19)	4.091 ± 7.6783 (20)	2.3288 ± 2.1584 (20)	0.094	0.271
Ruminococcus	0.0075 ± 0.0089 (6)	0.0036 ± 0.0027 (7)	0.0018 ± 0.0023 (3)	0.575	0.745
Saccharopolyspora	0.0014 ± 0 (1)	0.025 ± 0 (1)	0.0003 ± 0.0001 (2)	0.259	0.451
Scardovia	0.0023 ± 0.0028 (6)	0.016 ± 0.0289 (4)	0.0017 ± 0.0013 (6)	0.717	0.832
Schwartzia	0.0084 ± 0.0229 (15)	0.0046 ± 0.0065 (20)	0.0062 ± 0.0056 (18)	0.072	0.217
Selenomonas	0.341 ± 0.5316 (19)	0.1745 ± 0.2039 (20)	0.1903 ± 0.2269 (20)	0.822	0.865
Sharpea	0.0067 ± 0.0104 (9)	0.0058 ± 0.0068 (13)	0.0066 ± 0.0103 (11)	0.905	0.914
SHD.231	0.0011 ± 0.0007 (6)	0.0019 ± 0.0034 (16)	0.0006 ± 0.0007 (11)	0.034	0.156
Shuttleworthia	0.0099 ± 0.0162 (13)	0.0063 ± 0.0075 (20)	0.0052 ± 0.0039 (19)	0.769	0.857
Sneathia	0.1366 ± 0.1084 (2)	0.0003 ± 0.0001 (2)	0.0008 ± 0.0006 (2)	0.102	0.286
Sphaerochaeta	0.0016 ± 0.0021 (4)	0.0092 ± 0.0151 (4)	0.0022 ± 0.0025 (4)	0.841	0.876
Sporanaerobacter	0.0115 ± 0 (1)	0.0097 ± 0.0113 (6)	0.0091 ± 0.0097 (6)	0.962	0.962
Sporotomaculum	0.0008 ± 0.0007 (8)	0.0006 ± 0.0003 (6)	0.0005 ± 0.0002 (5)	0.646	0.81
Staphylococcus	0.0237 ± 0.0202 (19)	0.018 ± 0.0433 (20)	0.0046 ± 0.0049 (20)	0.001	0.017
Streptococcus	11.845 ± 6.8173 (19)	12.2051 ± 6.0074 (20)	16.5209 ± 11.1796 (20)	0.473	0.673

Table 3 (continued)

Genera	Comparison of genera across the three groups			Statistical significance (*p* value)	Hochberg-FDR (*p* value)
	Mean ± SD (*n*)				
	CP	**HTC**	**HI**		
Streptomyces	0.0039 ± 0.0028 (19)	0.0023 ± 0.0025 (19)	0.0052 ± 0.0077 (20)	0.063	0.211
Syntrophomonas	0.0045 ± 0.0044 (17)	0.0037 ± 0.0041 (15)	0.0027 ± 0.0034 (17)	0.227	0.441
Tannerella	0.0226 ± 0.025 (17)	0.0482 ± 0.0483 (20)	0.0594 ± 0.0889 (20)	0.066	0.211
Tissierella_Soehngenia	0.0008 ± 0.001 (12)	0.0008 ± 0.0001 (2)	0.0004 ± 0.0003 (4)	0.198	0.4
Treponema	0.2993 ± 1.0812 (19)	0.3583 ± 0.4243 (20)	0.5129 ± 1.141 (20)	0.023	0.122
Trichococcus	0.0005 ± 0.0003 (3)	0.001 ± 0 (1)	0.0003 ± 0.0001 (2)	0.304	0.505
Veillonella	1.8563 ± 2.2104 (19)	1.9628 ± 2.0189 (20)	1.9465 ± 1.8309 (20)	0.782	0.858

n number of participants

Functional characteristics of the oral bacteriome

Greengenes (gg13) and KEGG (Kyoto Encyclopedia of Gene and Genomics) databases were used as a reference database (Langille et al. 2013) while analyzing the functional characteristics using the PICRUSt software. The OTUs having 97% identity were selected for the PICRUSt analysis. The overall analysis revealed that the majority of the OTUs (47.38%) showed functional characteristics with the genes involved in metabolic pathways followed by the genes of genetic information/processing (21.71%). The percentage OTUs involved with environmental information processing and cellular processes were 12.67% and 1.97%, respectively, followed by 0.924% OTUs correlated with the genes of human diseases at the tier I level (Supplemental Table S9). Tier II level analysis provided better resolution of distribution of genes where a major portion of genes was involved in membrane transport (11.20%) and in replication/repair (10.01%) followed by genus involved in the metabolism of carbohydrate (9.62%) and amino acid (9.14%) (Supplemental Table S10). The tier III level analysis further elaborated the distribution of genes that included transporter (5.527%), DNA repair and recombination proteins (3.23%), and ABC transporters (3.07%) (Supplemental Table S11). Comparative analysis through the STAMP tool revealed an elevated number of genes (Benjamin–Hochberg FDR-adjusted value) involved in DNA repair and recombination, bacterial chemotaxis, bacterial motility, PPAR (peroxisome proliferator-activated receptor) signaling pathway, lysozyme, prostate cancer, cellular antigens, and pathways in cancer among CP group participants as compared to the HTC group (Supplemental Table S12; Fig. 4).

Discussion

The SLT burden in India accounts for 199.4 million adults where the contribution of males (141.2 million) is almost double the female (58.2 million) users (GATS-2, GoI). To the best of our knowledge, there is no report that discusses the oral bacteriome composition of SLT users having OSCC. Therefore, an attempt was made to understand the bacterial

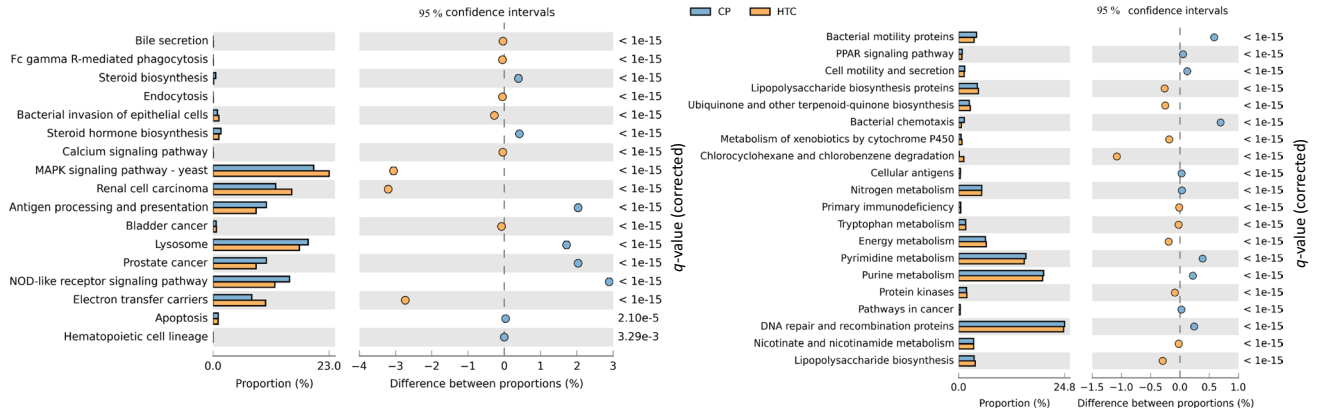


Fig. 4 The PICRUSt analysis at tier III reveals significant differences among the genes in between HTC and CP group participants

composition of SLT users who suffered from OSCC. Overall, a total of 18 phyla and 166 genera were assigned among the CP group, where 107 genera were common in all three groups (HI, HTC, and CP). Smokeless tobacco harbors a heavy microbial load and may introduce several environmental bacteria into the oral cavity (Han et al. 2016; Smyth et al. 2017). The altered tongue microbiome of smokeless tobacco users identified acetaldehyde (a potential carcinogen) producing bacteria (Halboub et al. 2020). In the present investigation, the prevalence of such bacteria (*Neisseria*, *Rothia*, *Prevotella*, and *Streptococcus*) was non-significantly different (p value > 0.05); however, their mean relative abundance (%) was comparatively higher in the CP groups as compared to their control groups (HI and HTC).

Statistical-level analysis revealed three significantly different phyla and 14 genera among the groups, where *Fusobacteria* (p value 0.01) was more associated with the CP group, while the dominance of *Spirochaetes* (p value 0.018) was almost similar in the HTC group. It indicates the strong association of *Fusobacteria* spp. with the oral cavity of the smokeless tobacco users. *Fusobacteria* spp. are well known for their role in several oral diseases (Bashir et al. 2016; Surlin et al. 2020) including oral cancer (Harrandah et al. 2020; Fujiwara et al. 2020). *Chloroflexi* (p value 0.05) was more associated with the HI group which indicates its association with the healthy oral cavity.

Of the 14 significantly different genera, nine genera (*Catonella*, *Coprococcus*, *Enterococcus*, *Fusobacterium*, *Gordonia*, *Lactobacillus*, *Microbacterium*, *Moryella*, and *Oribacterium*) had already been reported as significantly different among the HTC group in comparison with HI group participants (Srivastava et al. 2021). Therefore, smokeless tobacco exhibits potential to alter the bacterial diversity in the SLT users. Besides, it may be hypothesized that these bacteria might be positively correlated with smokeless tobacco products. *Microbacterium* (Hung et al. 2011; Tsuzukibashi et al. 2015), *Gordonia* (Hung et al. 2011), and *Enterococcus* (Zehnder and Guggenheim 2009; Komiyama et al. 2016) have been reported as transient and underrepresented bacteria of the human oral cavity. These bacteria must be carefully examined for their growth in the presence of nicotine or tobacco extracts for their role in nitrate reduction for assisting tobacco-specific nitrosamine formation. The dominance of *Moryella* (Costalonga and Herzberg 2014), *Enterococcus* (Komiyama et al. 2016), and *Staphylococcus* (Chocolatewala et al. 2010) like saccharolytic bacteria among the CP group further create an acidic microenvironment of solid tumors (Chen et al. 2015). However, the dominance of *Oribacterium* (Guerrero-Preston et al. 2016; Lim et al. 2018) and *Enterococcus* (Lee et al. 2017) had already been reported in oral cancers.

The antecedent biofilm-forming bacteria *Streptococcus* was comparatively lower in abundance in the CP group as compared to the HI and HTC group that corroborates the findings of Schmidt et al. (2014) and Zhang et al. (2020). *Streptococcus* spp. are considered as health commensal bacteria that produce bacteriocin and H_2O_2 to control the growth of several cariogenic pathogens (Sintim and Gursoy 2016; Lopez-López et al. 2017), while the prevalence of *Staphylococcus* spp. among the CP group opens the discussion in several facets. Some reports emphasize the abundance of methicillin-resistant *Staphylococcus aureus* in the oral cavity of the cancer patients either during the surgery or post-surgery incubation (Miyake et al. 2007; Pan et al. 2014). *Staphylococcus* spp. are also observed as inhabitant bacteria of various SLT products (Han et al. 2016; Smyth et al. 2017; Srivastav et al. 2020), and this could be the reason for their abundance in the oral cavity of CP group who chew tobacco regularly. *Staphylococcus* spp. bind with the inert oral surfaces by microbial surface components recognizing adhesive matrix molecules (MSCRAMMs) and participate in biofilm formation (Yong et al. 2019). Therefore, it may invite other gram-negative bacteria such as *Fusobacterium* and *Porphyromonas* during the biofilm formation and lead to the onset of oral diseases in a very similar manner as *Streptococcus* does (Jenkinson and Lamont 2005). We also found *Fusobacterium* as a significantly abundant genus in the CP group that has been reported for proliferation and progression of oral cancer (Bashir et al. 2016; Surlin et al. 2020; Harrandah et al. 2020; Fujiwara et al. 2020). Zhang et al. (2020) have recently identified that the cancer tissues are enriched with the *Fusobacterium* spp. The dominance of *Campylobacter* among CP group participants is an alarming situation in the context of oral cancer as in a few reports, a trio of *Fusobacterium-Campylobacter-Leptotrichia* has been reported as abundant genera on cancer tissues (Warren et al. 2013; Amer et al. 2017). Amer et al. (2017) also emphasized that co-occurrence of *Fusobacterium* and *Campylobacter* might induce tumorigenesis by activating the IL-6-STAT3 axis and β-catenin signaling which enhances the expression of the proinflammatory *IL-8* gene (Rubinstein et al. 2013). *Campylobacter rectus* has been proven for its role in an oral abscess (Mahlen and Clarridge 2009; Liu et al. 2016). Though in the present investigation, *Leptotrichia* was not identified as a significantly dominant genus, however, at mean level abundance, it was comparatively higher among the CP group. Al-Hebshi et al. (2017) have also reported *F. nucleatum* and *Campylobacter* sp. oral taxon as populated bacteria among OSCC patients. Members of *Capnocytophaga* spp. constitute the normal microflora of the human oral cavity; however, it causes several inflammatory diseases among immunocompromised and immunocompetent humans (Antezack et al. 2021). *Capnocytophaga gingivalis* has also been reported in cancer development, as it showed a strong correlation with OSCC (Mager et al. 2005; Karpiński 2019; Chocolatewala et al. 2010). The saliva of lung cancer patients showed

enrichment with the *Capnocytophaga* spp. (Yan et al. 2015). It would be interesting to know the mechanism of carcinogenesis due to *Capnocytophaga*, a potential biomarker of oral cancer (Mager et al. 2005). *Capnocytophaga*, *Fusobacterium*, and *Eikenella* showed significantly higher abundance in the oral cavity of the OSCC patients (Zhao et al. 2017) and support the present findings. *Eikenella* had been characterized as gram-negative and facultative anaerobic proteobacteria. A few reports suggest the significant abundance of *Eikenella* in oral cancer patients (Zhao et al. 2017; Chattopadhyay et al. 2019). Chattopadhyay et al. (2019) observed that *Eikenella corrodens* was significantly abundant in the oral cancer participants. Members of the genus *E. corrodens* showed their abundance in colorectal cancer (Allali et al. 2015). Yang et al. (2018) also observed the prevalence of *E. corrodens* among OSCC stage 4 participants along with *Neisseria elongata*, *Oribacterium* spp., and *Dialister pneumosintes*. The genus must be carefully studied for its etiology of oral cancer. In the present investigation, *Corynebacterium* was also detected as a significantly different genus that was more associated with the HTC group. Biogeography of a human oral microbiome revealed that *Corynebacterium* spp. preferred sub- and supra-gingival sites of the oral cavity similar to *Eikenella* and participate in dental plaque formation (Mark Welch et al. 2016). Furthermore, an abundance of *Capnocytophaga* was also significantly higher in plaque sites along with *Corynebacterium*, *Lautropia*, and *Rothia* (Mark Welch et al. 2016). It would be interesting to identify the presence of *Corynebacterium* spp. in oral cancer as many studies suggest their prevalence in cancer tissues (Fisher et al. 2012; Mukherjee et al. 2017). Squamous cell carcinoma of the oral tongue (OMTC) showed an elevated count of *Corynebacterium* along with *Enterococcus*, *Streptococcus*, *Actinomyces*, *Micrococcus*, and *Rothia mucilaginosa* (Mukherjee et al. 2017). *Corynebacterium* spp. had also been identified as an inhabitant bacterium of smokeless tobacco products (Tyx et al. 2016; Chopyk et al. 2017; Srivastav et al. 2020).

Overall, the dominance of cariogenic bacteria such as *Staphylococcus*, *Fusobacterium*, and *Corynebacterium* may be considered as the major players for tumorigenesis proliferation among the CP group here, as these bacteria have been well documented for their direct involvement in nitrate/nitrite reduction that may assist in TSNA formation (Di Giacomo et al. 2007; Fisher et al. 2012; Hyde et al. 2014).

The PICRUSt-based STAMP analysis revealed a greater number of genes involved in DNA repair and recombination proteins that were also reflected in the elevated count of genes engaged in purine and pyrimidine metabolism among the CP group as compared to the HTC group. Un-repaired DNA either directs the cells to apoptosis or may stimulate the mutation for initiating carcinogenesis (De Zio et al. 2013). Smokeless tobacco imposes a potential risk factor to mutate the DNA, where TSNA lead to impairment of DNA by exerting a mutagenic effect on DNA (Swann and Magee 1968; de la Monte and Tong 2009) by alkylating *N*-7 of guanine and led to breakage of DNA (Otteneder and Lutz 1999). Moreover, activated nitrosamine also damages DNA by generating reactive oxygen species and hydrogen peroxides (Espey et al. 2002; Chuang and Hu 2006). A higher proportion of genes were observed in nitrogen metabolism similar to the HTC group of our previous investigation (Srivastava et al. 2021) that indicate abundant nitrite and nitrosamine levels which may react with nicotine to form TSNA-like carcinogens (Bush et al. 2001). In contrast, genes involved in nicotinate/nicotinamide degradation were almost similar in both the groups and suggest the bacterial-derived transformation of nicotinic acid from nicotinamide degradation (Hughes and Williamson 1953; Gadd and Johnson 1974). A greater number of genes were also involved in the metabolism of xenobiotic compounds, secondary metabolite degradation, chlorocyclohexane, and chlorobenzene degradation as expected. It advocates the involvement of bacteria to deal with exogenous environmental compounds. These complex compounds may either be introduced into the oral cavity from continuous tobacco chewing or derived as an intermediate from bacterial activity. Several studies show the influence of nutritional intake on the oral microbiome (Wade 2013; Kato et al. 2017; Murtaza et al. 2019). Kato et al. (2017) found a positive correlation in the relative abundance of *Betaproteobacteria* and *Fusobacteria* due to short-chain fatty acids intake while vitamin C supported the *Fusobacteria* spp. Similarly, a significant increase has been observed in *Streptococcus mutans* due to high sugar intake (Touger-Decker and van Loveren 2003). Zhang et al. (2020) emphasized lipopolysaccharide (LPS)-induced inflammation and its role in the progression and migration of OSCC. However, such genes were more associated with SLT users in the present investigation as compared to the CP group and therefore need to be carefully studied for their factual association with inflammation and tumorigenesis. Moreover, a higher proportion of genes involved in bacterial motility and chemotaxis among the CP group supported the hypothesis of Al-Hebshi et al. (2017) and Zhang et al. (2020), where such attributes might be associated with cancer-related inflammation. Besides, the elevated gene count of prostate cancer, antigen processing and presentation (Jhunjhunwala et al. 2021), pathways in cancer, Nod-like receptor (Saxena and Yeretssian 2014), signaling pathways (Zhao et al. 2021), and peroxisome proliferator-activated receptor (PPAR) signaling (Tachibana et al. 2008; Apostoli et al. 2014) among the CP group had already been documented in tumor progression. Therefore, these genes may be employed among healthy SLT users for determining the tumor risks and oral health.

Our results elucidate that the oral bacterial composition of OSCC patients varied significantly in the presence of smokeless tobacco from their control groups, i.e., HI and HTC groups (Fig. 3C). Overall, we observed that the oral cavity of CP group

participants harbors a greater number of bacteria involved in nitrate or nitrite reduction. These activities may be taking place due to SLT consumption that showed the prevalence of such bacteria. Thus, TSNAs could be considered as the foremost carcinogen for OSCC induction among this set of participants.

The functional attributes further support the hypothesis where an abundance of several genes may find a role in oral oncogenesis. However, alteration of these genes must be verified in various genera observed here as statistically different. This is preliminary information on the oral bacteriome of OSCC patients that must be extensively studied with a far greater number of participants by including various socio-economic factors. It would be further interesting to identify smokeless tobacco inhabitant bacteria and to link them with the oral bacteriome of OSCC patients. Such studies are highly required especially from South and Southeast Asian regions, where OSCC patients, as well as SLT users, are prevalent over the other regions of the world.

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Author contribution DV and PKG conceived the idea; DV designed experiments. DV and AS performed experiments. DV, AS, and SM analyzed the data. DV, AS, and AKD, SVSD contributed for reagents/samples/analysis tools. DV, PKG, and AS compiled the manuscript. DV, AS, and AKD edited the manuscript. All authors approved the final version of the manuscript.

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Data availability All Illumina MiSeq data in this study have been submitted to the NCBI Sequence Read Archive (SRA) database under the BioProject accession number PRJNA756784 (https://www.ncbi.nlm.nih.gov/sra/PRJNA756784).

Code availability Not applicable.

Declarations

Ethics approval The protocol of the study was approved by the Institutional Ethical Committee, Netaji Subhas Institute of Technology, New Delhi as well as All India Institute of Medical Sciences (AIIMS), New Delhi (Ref No. IEC/NP 166/2014/ RP-14/2014).

Consent to participate A signed consent was taken from all the participants in this investigation.

Conflict of interest The authors declare no competing interests.

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